Modulo Operation and Caesar Cipher

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Due Apr 7, 11:59 PM IST **Attempts** 3 every 8 hours

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Review Learning Objectives

1. Let's use a modified Caesar Cipher where c = (p+2x) mod 26, where c and p are the ciphertext and the plaintext, respectively, and x is the key. How many distinct keys, producing distinct encryption/decryption transformations, are there now? **4 / 4 points**

13

Correct
See Modulo Operation lesson and interpret the equation.

2. For c = (p+2x) mod 26, where c, p, and x are the ciphertext, the plaintext, and the key, respectively, what is the corresponding decryption? Select all that works. **4 / 4 points**

- [ ] p = (c-x) mod 26
- [x] p = (c+26-2x) mod 26

  Correct
  See Modulo Operation

- [ ] p = (c-x) mod 13
- [x] p = (c-2x) mod 26

  Correct
  See Modulo Operation lesson.

- [ ] p = (c-2x) mod 13

3. Let's now use a modified Caesar Cipher where c = (p+3x) mod 26, where c and p are the ciphertext and the plaintext, respectively, and x is the key. How many distinct keys, producing distinct encryption/decryption transformations, are there now? **4 / 4 points**

26

Correct
See Modulo Operation lesson and interpret the equation.